Mathematics: analysis and approaches
Standard level
Paper 2 — markscheme
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 1 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P2–01

Student name

Paper 2

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[27 marks]**.

W4ss Academy 2026

Section A

Question	Marks	Markscheme
1.	**[4]**	$\frac{\sum_{i=1}^{16} x_i}{16} = 14.5$ **(M1)** **Note:** Award **M1** for use of $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$. $\Rightarrow \sum_{i=1}^{16} x_i = 232$ **(A1)** new $\bar{x} = \frac{232 - 9}{15}$ **(A1)** $= 14.9$ $(= 14.8\overline{6}, = \frac{223}{15})$ **A1** **Note:** Do not accept 15.
2. (a)	**[2]**	**EITHER** recognising that half the total frequency is 10 (may be seen in an ordered list or indicated on the frequency table) ***(A1)*** **OR** $5 + 1 + 4 = 3 + x$ ***(A1)*** **OR** $\sum f = 20$ ***(A1)*** **THEN** $x = 7$ ***A1***
2. (b)	**[2]**	**METHOD 1** 1.58429 … 1.58 ***A2*** **METHOD 2** **EITHER** $\sigma^2 = \frac{5 \times (2 - 4.3)^2 + 1 \times (3 - 4.3)^2 + 4 \times (4 - 4.3)^2 + 3 \times (5 - 4.3)^2 + 7 \times (6 - 4.3)^2}{20}$ $(= 2.51)$ ***(A1)*** **OR** $\sigma^2 = \frac{5 \times 2^2 + 1 \times 3^2 + 4 \times 4^2 + 3 \times 5^2 + 7 \times 6^2}{20} - 4.3^2$ $(= 2.51)$ ***(A1)*** **THEN** $\sigma = \sqrt{2.51} = 1.58429 \ldots$ $= 1.58$ ***A1***

3. (a)	**[2]**	**EITHER** $P(S) + P(T) + P(S' \cap T') - P(S \cap T) = 1$ OR $P(S \cup T) = P((S' \cap T')')$ ***(M1)*** $0.7 + 0.2 + 0.18 - P(S \cap T) = 1$ OR $P(S \cup T) = 1 - 0.18$ **OR** a clearly labelled Venn diagram ***(M1)*** **THEN** $P(S \cap T) = 0.08$ (accept 8%) ***A1*** **Note:** To obtain the ***M1*** for the Venn diagram all labels must be correct and in the correct sections. For example, do not accept 0.7 in the area corresponding to $S \cap T'$.
3. (b)	**[2]**	**EITHER** $P(T \cap S') = P(T) - P(T \cap S)$ ($= 0.2 - 0.08$) OR $P(T \cap S') = P(T \cup S) - P(S)$ ($= 0.82 - 0.7$) ***(M1)*** **OR** a clearly labelled Venn diagram including $P(S)$, $P(T)$ and $P(S \cap T)$ ***(M1)*** **THEN** $= 0.12$ (accept 12%) ***A1***
3. (c)	**[2]**	$P(G \cap T) = P(T / G)P(G)$ (0.25×0.48) ***(M1)*** $= 0.12$ ***A1***
3. (d)	**[2]**	**METHOD 1** $P(G) \times P(T)$ ($= 0.48 \times 0.2$) $= 0.096$ ***A1*** $P(G) \times P(T) \neq P(G \cap T) \Rightarrow$ G and T are not independent ***R1*** **METHOD 2** $P(T \mid G) = 0.25$ ***A1*** $P(T \mid G) \neq P(T) \Rightarrow$ G and T are not independent ***R1*** **Note:** Do not award ***A0R1***.

Section B

Question	Marks	Markscheme
4. (a)	**[1]**	Stratified sampling ***A1***

4. (b)(i)	**[3]**	There are 260 students in total ***A1*** $\frac{84}{260} \times 9 = 2.91$ ***M1A1*** So 3 students will be selected. ***AG***
4. (b)(ii)	**[2]**	grade 9 $= \frac{60}{260} \times 9 \approx 2$, grade 10 $= \frac{83}{260} \times 9 \approx 3$, grade 11 $= \frac{33}{260} \times 9 \approx 1$ ***A2***
4. (c)	**[1]**	Systematic sampling ***A1***
4. (d)(i)	**[2]**	$r = -0.901$ ***A2***
4. (d)(ii)	**[1]**	The negative value of r indicates that more time spent on social media leads to lower self-esteem, supporting the principal's concerns. ***R1***
4. (d)(iii)	**[1]**	r being close to –1 indicates there is strong correlation, so a regression line is appropriate. ***R1***